$$= 125x + 5x^2 - \frac{x^3}{27} + k$$

Fixed cost k= 250

$$C = 125x + 5x^2 - \frac{x^3}{27} + 250$$

When x = 15

$$C = 125(15) + 5(15)^2 - \frac{(15)^3}{27} + 250$$

$$= 1875 + 1125 - 125 + 250$$

C = ₹3,125

Example 3.11

The marginal cost function $MC = 2 + 5e^x$ (i) Find C, if $C(0)=100$ (ii) Find AC.

Solution:

Given $MC = 2 + 5e^x$

$$C = \int MC\, dx + k$$

$$= \int (2 + 5e^x)\, dx + k$$

$$= 2x + 5e^x + k$$

$x = 0 \Rightarrow C = 100$,

$100 = 2(0) + 5(e^0) + k$

$\boxed{k = 95}$

$C = 2x + 5e^x + 95$.

Average cost $= \frac{C}{x} = \frac{2x + 5e^x + 95}{x}$

$AC = 2 + \frac{5e^x}{x} + \frac{95}{x}$.

Rate of growth or sale

If the rate of growth or sale of a function is a known function of t say $f(t)$ where t is a time measure, then total growth (or) sale of a product over a time period t is given by,

$$\text{Total sale} = \int_0^r f(t)\, dt,\ 0 \le t \le r$$

Example 3.12

The rate of new product is given by $f(x) = 100 - 90\, e^{-x}$ where x is the number of days the product is on the market. Find the total sale during the first four days. (e^{-4}=0.018)

Solution:

Total sale $= \int_0^4 (100 - 90e^{-x})\, dx$

$$= (100x + 90e^{-x})_0^4$$

$$= 400 + 90e^{-4} - (0 + 90)$$

$$= 400 + 90(0.018) - 90$$

= 311.62 units

Example 3.13

A company produces 50,000 units per week with 200 workers. The rate of change of productions with respect to the change in the number of additional labour x is represented as $300 - 5x^{2/3}$. If 64 additional labours are employed, find out the additional number of units, the company can produce.

Solution:

Let p be the additional product produced for additional of x labour,

$$\frac{dp}{dx} = 300 - 5x^{2/3}$$

$$p = \int_0^{64} \left(300 - 5x^{\frac{2}{3}}\right) dx$$

$$= \left[300x - 3x^{\frac{5}{3}}\right]_0^{64}$$

$$= 300 \times 64 - 3(64)^{5/3}$$

$$= 16128$$

$\therefore$ The number of additional units produced 16128.

Total number of units produced by 264 workers = 50,000 + 16,128 = 66128 units.

Example 3.14

The rate of change of sales of a company after an advertisement campaign is represented as, $f(t) = 3000e^{-0.3t}$ where t represents the number of months after the advertisement.

XII Std - Business Maths & Stat EM Chapter 3.indd 66 2/28/2020 5:49:02 PM